

Application of Statistical Physics to Complex Systems (SC478)

1 Course Placement

Application of Statistics to Complex Systems can be a science elective course for students taking a B.Tech (Honours) degree in ICT and an elective for MnC students.

2 Course Format

The course will consist of 3 hours of lectures each week.

3 Course Content

Concepts of statistics have proved to be highly useful in probing the behaviour of different physical systems, typically consisting of a large number of particles. These were traditionally first developed in the context of thermal physics, but have subsequently been used in other diverse areas. This course is intended to introduce probabilistic approaches to studying systems with large number of degrees of freedom. These systems can range from ideal gases, which consist of large numbers of particles to modeling the dynamics of the stock market using unifying concepts such as entropy and ensembles. This course will introduce both the formalism needed for systematically studying such systems and many concrete applications wherein the techniques are applied.

4 Textbooks and References

1. *Concepts in Thermal Physics*, S. Blundell and K. Blundell, Oxford University Press, 2006,
2. *An Introduction to Thermal Physics*, D. V. Schroeder, Oxford University Press, 2021,
3. *Fundamentals of Statistical and Thermal Physics*, F. Reif, , McGraw-Hill Book Company, New York.

Other reference books/material will be suggested in class from time to time, as appropriate.

5 Grading Policy

Grades will be based on performance in one mid-semester/in-semester examination and one end-semester exam. The weightage will be 35% and 65% respectively.

6 Course Outcomes

Students will learn to formulate a physical problem and solve it, using analytical and computational methods. Problem-solving will be the main focus of this course. After completion, the students should be able to,

- To understand and analyse the physical world, using a basic set of principles.
- To design, model and investigate complex engineering problems.

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
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7 Course material and Tentative Lecture Schedule

7.1 Concepts of Probability

Time Allotted (10 Lectures)

Theory (5 lectures): General definitions and statistical terminology, Random variables, Probability Distributions, Probability distributions for large n , Information and Entropy, Random Walk and fat-tailed distributions.

Applications (in brief) (5 lectures): Using random variables as a tool to study physical systems. Sampling from Probability Distributions, Monte Carlo Integration and applications in simple systems.

7.2 Statistical Formulation of a Physical Problem

Time Allotted (6 Lectures)

How to specify state of a system, Statistical Ensemble, Density of states.

7.3 Concepts of Thermodynamics

Time Allotted (7 Lectures)

Zeroth Law, First Law and Second Law of Thermodynamics, Entropy, Concept of equilibrium and thermodynamic potentials, Uses of the above concepts in areas such as finance and the stock market. Maximum entropy principle in complex systems.

7.4 Classical Statistical Mechanics

Time Allotted (17 Lectures)

Theory (10 Lectures): Microcanonical, Canonical and Grand Canonical Ensembles, Partition Function, Ising model and its uses in physics and in other areas such as the stock market, Exact solution of the Ising model. Applications (in brief) (3 Lectures): Metropolis-Hastings algorithm for Monte Carlo simulation of the Ising model and calculating different physical quantities in this context. Scaling and its applicability in complex networks.

Note: The number of lectures assigned for each topic is tentative and would ultimately depend on the progress of students. The list of topics is indicative of the basic material to be covered and may change slightly, depending on progress.

8 Objectives and Learning Outcomes

The objectives of this course is to teach the students how to come up with a mathematical formulation of a physical problem and then select appropriate methods to solve it. Laboratory sessions will mainly be used to supplement the lectures, giving students an opportunity to put theory into practice.

9 Course Credits

3.0 (3 – 0 – 0 – 3)